BEFORE YOU READ

Twelve derivatives in five families

Every derivative on this page belongs to one of five families. Learn the family and the members follow.

- L0 **Common derivatives**
  - L1 **Constant**
    - L2 **c** — 0
  - L1 **Power**
    - L2 **x^n** $n \in \mathbb{R}$ — $n \cdot x^{n-1}$
  - L1 **Trigonometric**
    - L2 **sin x** — $\cos x$
    - L2 **cos x** — $-\sin x$
    - L2 **tan x** — $\sec^2 x$
    - L2 **cot x** — $-\csc^2 x$
    - L2 **sec x** — $\sec x \cdot \tan x$
    - L2 **csc x** — $-\csc x \cdot \cot x$
  - L1 **Exponential**
    - L2 **e^x** — e^x
    - L2 **a^x** $a > 0, a \neq 1$ — $a^x \cdot \ln a$
  - L1 **Logarithmic**
    - L2 **ln x** $x > 0$ — $1 / x$
    - L2 **$\log_a x$** $a > 0, a \neq 1$ — $1 / (x \cdot \ln a)$

TAKEAWAY

The building blocks

These twelve are the starting points; the **derivative rules** combine them into the derivative of almost any function.